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Roll No. | | | | | | | |

Mid Semester Examination Oct 2024

Name of the Program: B.Tech *Semester: III*

Name of the Course: Electronics & Communication Engineering *Course Code.* TEC-301

Paper Name: Electronic Devices

Time: 1:30 Hours

MM: 50

Note:
(i) Answer **all the questions** by choosing ***any one of the sub questions***.
(ii) Each question contains two parts a & b. Attempt any one part of choice from each question.

Q.1	**(10 Marks)**	CO1
a)	Explain energy band diagram for P type and N type semiconductors. Explain the V-I characteristic curve of P-N junction diode in forward and reverse bias condition.	
	OR	
b)	Derive the expression for Potential barrier in p-n junction. The electron mobility in silicon (Si) is 1000 $cm^2/(V\text{-}s)$. What is the electron drift velocity if the voltage across the sample is 10 V and the sample is d=1 mm thick?	
Q.2	**(10 Marks)**	CO1
a)	Explain Drift Current Diffusion Current Mobility Resistivity	
	OR	
b)	Write differences between P type and N type semiconductor. Derive the expression for electric field in p-n junction.	
Q.3	**(10 Marks)**	CO1, CO2
a)	Explain the formation of depletion region in PN junction Diode. An abrupt p-n junction has $10^{15}/cm^3$acceptor atom concentration on p side and $3\times 10^{18}/cm^3$ donor atom concentration on other side. Intrinsic concentration at 300K is $5\times 10^{11}/cm^3$. Calculate space charge width of depletion layer in n-side and p-side at 300 K for relative permittivity of 11.	
	OR	

b)	Explain in detail (i) Light Emitting Diode (ii) Schottky barrier diode	
Q.4	**(10 Marks)**	CO1, CO2
a)	Explain P-N Junction Capacitance. Draw the small signal model of diode for high frequency signal. Explain the working of Tunnel diode and tunneling effect.	
	OR	
b)	Calculate the diffusion current density when the concentration of electron varies from the $1 * 10^{18}$ to $7 * 10^{17}$ cm^{-3} over a distance of 0.10 cm. $D = 225 cm^2/s$. In a semiconductor, the electron concentration n is $10^{16} cm^{-3}$. the electron mobility μ_n is $1350 cm^2/V\text{-}s$, and an electric field E of 1000V/cm is applied. Calculate the drift current density due to electrons.	
Q.5	**(10 Marks)**	CO2
a)	Explain the working of Zener diode. Explain Zener and Avalanche breakdown.	
	OR	
b)	Write down the differences between normal p-n junction diode and Schottky barrier diode. A p-n junction diode made of Si has $NA=10^{19}/cm^3$ and $ND=10^{14}/cm^3$. Intrinsic concentration at 300K is $5\times 10^{12}/cm^3$Calculate the built-in potential of diode at 300K. The relative permittivity of Si is 11.8	